

Fernando Lucatelli Nunes

Category Theory • Programming Languages • Formal Methods

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ORCID 0000-0002-1817-2797 • [arXiv](#) • [Google Scholar](#) • [Scopus 57188838707](#) • [Mathematics Genealogy](#)

I am a mathematician and computer scientist working across category theory, programming-language semantics, and formal methods. My research develops low- and higher-dimensional categorical structures—particularly through two- and three-dimensional universal algebra—and uses them to obtain principled accounts of program transformations, automatic differentiation, recursion and iteration, algebraic effects, and probabilistic computation. Across these areas, I seek structural results that are mathematically intrinsic, technically exact, and useful in the design of reliable computational systems. I hold a PhD in Mathematics (2018); my second PhD, in Computer Science at Utrecht University, has been submitted and its public defence is scheduled for September 2026.

ACADEMIC POSITIONS AND AFFILIATIONS

03/2026–present	Assistant Professor , CMUC , Department of Mathematics, University of Coimbra, 3000-143 Coimbra, Portugal.
01/2026–03/2026	Research Fellow , School of Computer Science, University of Birmingham, United Kingdom; DFG project HOMBRé.
2025	Postdoctoral Fellow , Department of Information and Computing Sciences, Utrecht University, The Netherlands.
11/2020–11/2025	Doctoral Researcher , Department of Information and Computing Sciences, Utrecht University, The Netherlands.
2023	Postdoctoral Fellow , Department of Mathematics and Statistics, York University, Canada.
2023	Fields Research Fellow , Fields Institute for Research in Mathematical Sciences, Canada.
2022	Oberwolfach Leibniz Fellow , Mathematisches Forschungsinstitut Oberwolfach, Germany.
2018–2020	Research Fellow , Department of Mathematics, University of Coimbra, Portugal.
2018–2019	Invited Assistant Professor , Department of Mathematics, University of Coimbra, Portugal.
2018	Postdoctoral Fellow , University of Coimbra, Portugal.
2012–2018	Doctoral Researcher , Department of Mathematics, University of Coimbra, Portugal.

VISITING ACADEMIC APPOINTMENTS

2026–present	Visiting Researcher , Utrecht University , The Netherlands.
2025	Visiting Researcher, IMPA, Rio de Janeiro, Brazil.
2018, 2023–2025	Visiting Researcher, UCLouvain, Belgium.
2019, 2025	Visiting Researcher, University of Cape Town, South Africa.
2019, 2025	Visiting Researcher, Stellenbosch University, South Africa.

OTHER RESEARCH AFFILIATIONS

2026	Research Fellow, Theory of Computation group, School of Computer Science, University of Birmingham, United Kingdom.
2020–2026	Researcher in Programming Language Semantics, Software Technology group, Utrecht University, The Netherlands.

RESEARCH PROFILE

In pure mathematics, my research develops the theory and applications of generalised categorical structures in low and higher dimensions, primarily through two- and three-dimensional universal algebra. This programme addresses intrinsic questions in categorical algebra, Grothendieck descent theory, categorical Galois theory, coherence, and presentations, while maintaining connections with proof theory, algebraic topology, homotopy theory, geometry, and logic.

In computer science, my research focuses on programming-language semantics, formal methods, and software technology, with particular emphasis on correctness-by-construction methodologies and principled program transformations. I use categorical semantics, logical relations, and type-theoretic structure to study automatic differentiation, recursion and iteration, algebraic effects and handlers, probabilistic programming, and other emerging programming paradigms.

Across these areas, I aim to lead a research programme that brings categorical and algebraic ideas to bear both on substantial problems in pure mathematics and on the foundations of modern computation. A central objective is to train researchers who can move confidently between intrinsically motivated mathematics and foundational as well as practical challenges in computer science.

EDUCATION

PHD IN MATHEMATICS – AWARDED 2018

10/2012–01/2018

Joint doctoral programme in Mathematics, Portugal.

- Thesis: *Pseudomonads and Descent* (2017; 209 pages).
- Supervisor: Prof. Maria Manuel Clementino.
- Public defence: 24 January 2018; awarded summa cum laude.
- Examination committee: Prof. Martin Hyland, Prof. Marco Mackaay, Prof. Lurdes Sousa, and Prof. Peter Beier Gothen.
- Keywords: pseudomonads; biadjoint triangle theorems; presentations and coherence; freely generated categorical structures; Grothendieck descent; categorical Galois theory; weak limits.

PHD IN COMPUTER SCIENCE – THESIS SUBMITTED; DEFENCE SCHEDULED

11/2020–09/2026

Utrecht University, Department of Information and Computing Sciences.

- Thesis: *Freely Generated Categorical Structures and Automatic Differentiation* (submitted 3 November 2025; 380 pages).
- Supervisors: Prof. Gabriele Keller and Dr. Matthijs Vákár.
- Public defence scheduled for 28 September 2026.
- Assessment committee: Prof. Ieke Moerdijk, Prof. Bart Jacobs, Prof. Sam Staton, Prof. Robin Cockett, and Prof. Patricia Johann.
- Keywords: denotational semantics; automatic differentiation; principled program transformations; recursion and iteration; categorical semantics; logical relations; type systems; Grothendieck constructions; extensive categories.

REFEREED JOURNAL PUBLICATIONS

- F. Lucatelli Nunes** and **M. Vákár**. **Monoidal closure of Grothendieck constructions via Σ -tractable monoidal structures and Dialectica formulas.** *Theory and Applications of Categories* 44(35), 1153–1217 (2025), 65 pages. [Journal](#); [arXiv](#).
- F. Lucatelli Nunes**, **R. Prezado**, and **M. Vákár**. **Free extensivity via distributivity.** *Portugaliae Mathematica* 82(1–2), 177–204 (2025), 28 pages. [DOI](#); [arXiv](#).
- R. Prezado** and **F. Lucatelli Nunes**. **Generalized multicategories: change-of-base, embedding, and descent.** *Applied Categorical Structures* 32, Article 35 (2024), 89 pages. [DOI](#); [arXiv](#).
- M. M. Clementino** and **F. Lucatelli Nunes**. **Lax comma 2-categories and admissible 2-functors.** *Theory and Applications of Categories* 40(6), 180–226 (2024), 47 pages. [DOI](#); [arXiv](#).
- M. M. Clementino**, **F. Lucatelli Nunes**, and **R. Prezado**. **Lax comma categories: cartesian closedness, extensivity, topologicity, and descent.** *Theory and Applications of Categories* 41(16), 516–530 (2024), 15 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes** and **M. Vákár**. **Automatic differentiation for ML-family languages: correctness via logical relations.** *Mathematical Structures in Computer Science* 34(8), 747–806 (2024), 60 pages. [DOI](#); [arXiv](#).

- R. Prezado and **F. Lucatelli Nunes**. **Descent for internal multicategory functors**. *Applied Categorical Structures* 31, Article 11 (2023), 18 pages. [DOI](#); [arXiv](#).
- M. M. Clementino and **F. Lucatelli Nunes**. **Lax comma categories of ordered sets**. *Quaestiones Mathematicae* 46(S1), 145–159 (2023), 15 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes** and M. Vákár. **CHAD for expressive total languages**. *Mathematical Structures in Computer Science* 33(4–5), 311–426 (2023), 116 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes**, R. Prezado, and L. Sousa. **Cauchy completeness, lax epimorphisms and effective descent for split fibrations**. *Bulletin of the Belgian Mathematical Society–Simon Stevin* 30(1), 130–139 (2023), 10 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes**. **Semantic factorization and descent**. *Applied Categorical Structures* 30(6), 1393–1433 (2022), 41 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes** and L. Sousa. **On lax epimorphisms and the associated factorization**. *Journal of Pure and Applied Algebra* 226(12), Article 107126 (2022), 27 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes**. **Descent data and absolute Kan extensions**. *Theory and Applications of Categories* 37(18), 530–561 (2021), 32 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes**. **Pseudoalgebras and non-canonical isomorphisms**. *Applied Categorical Structures* 27(1), 55–63 (2019), 9 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes**. **On lifting of biadjoints and lax algebras**. *Categories and General Algebraic Structures with Applications* 9(1), 29–58 (2018), 30 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes**. **Pseudo-Kan extensions and descent theory**. *Theory and Applications of Categories* 33(15), 390–444 (2018), 55 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes**. **On biadjoint triangles**. *Theory and Applications of Categories* 31(9), 217–256 (2016), 40 pages. [DOI](#); [arXiv](#).

ACCEPTED FOR PUBLICATION

- F. Lucatelli Nunes**, G. Plotkin, and M. Vákár. **Unraveling the iterative CHAD**. Accepted for publication in *Mathematical Structures in Computer Science*; 58 pages (2025). [arXiv:2505.15002](#).
- F. Lucatelli Nunes** and M. Vákár. **Free doubly-infinitary distributive categories are cartesian closed**. Accepted for publication in *Applied Categorical Structures*; 32 pages (2024; substantially revised 2026). [arXiv:2403.10447](#).

PREPRINTS

- F. Lucatelli Nunes** and W. Tholen. **From Grothendieck cofibrations to factorization systems: a formal 2-monadic account**. Preprint, 46 pages (2026); DMUC Preprint 26-43. [arXiv:2607.27541](#).
- F. Lucatelli Nunes**, D. Simm, and M. Vákár. **Simply typed reverse-mode automatic differentiation with variants: denotational correctness via idempotent completion**. Preprint, 55 pages (2026); DMUC Preprint 26-40. [arXiv:2607.14453](#).
- D. Simm, **F. Lucatelli Nunes**, and M. Vákár. **Backpropagation for effectful languages I: finite probability and discrete output algebraic effects**. Preprint, 71 pages (2026); submitted to POPL 2027; DMUC Preprint 26-35. [arXiv:2607.13935](#).
- F. Lucatelli Nunes** and R. Prezado. **Functors preserving effective descent morphisms**. Preprint, 17 pages (2024); DMUC Preprint 24-44. [arXiv:2410.22876](#).
- F. Lucatelli Nunes**. **Freely generated n -categories, inserter and presentations of low dimensional categories**. Preprint, originally 2017; fully revised in 2026, 76 pages; DMUC Preprint 17-20. [arXiv:1704.04474](#).

SELECTED ONGOING WORK

- P. B. Levy and **F. Lucatelli Nunes**. **Mac Lane strictification in place**. Manuscript in preparation (2026).
- F. Lucatelli Nunes**, G. Plotkin, and M. Vákár. **CHAD for recursion**. Manuscript in preparation.
- F. Lucatelli Nunes**, S. Goncharov, M. Vákár, and T. Van der Linden. **Symmetric infinitary monoidal categories**. Manuscript in preparation.
- F. Lucatelli Nunes** and M. M. Clementino. **On lax comma 2-categories**. Manuscript in preparation.
- F. Lucatelli Nunes** and W. Tholen. **Spanning the tale of monads and descent**. Manuscript in preparation.

OTHER RESEARCH OUTPUTS

- F. Lucatelli Nunes, G. Plotkin, and M. Vákár. **Homomorphic reverse differentiation of iteration**. Extended abstract, LAFI 2024 (Tenth Workshop on Languages for Inference at POPL), 2 pages. [Workshop page](#).
- F. Lucatelli Nunes and M. Vákár. **Logical relations for partial features and automatic differentiation correctness**. *Oberwolfach Preprints*, OWP-2023-09 (2023), 25 pages. [DOI](#); [arXiv](#).
- F. Lucatelli Nunes. **Freely generated categorical structures and automatic differentiation**. Computer Science PhD thesis, Utrecht University, submitted 2025, 380 pages.
- F. Lucatelli Nunes. **Pseudomonads and descent**. Mathematics PhD thesis, 2017; degree awarded 2018, 209 pages. [Repository](#); [Chapter 1 on arXiv](#).
- F. Lucatelli Nunes. **Espaços não reversíveis**. *Revista Matemática Universitária* 48–49, 22–26 (2012), 5 pages. [Article](#).

UNDERGRADUATE AND GRADUATE MONOGRAPHS

- F. Lucatelli Nunes. **Basic concepts of topology for topological dynamics**. In Portuguese, 2009, 116 pages.
- F. Lucatelli Nunes. **Basic topological dynamics and basic applications to number theory**. In Portuguese, 2009, 57 pages.

GRANTS, FELLOWSHIPS, AND FUNDED PROJECTS

SELECTED RESEARCH FELLOWSHIPS AS PRINCIPAL INVESTIGATOR

- 2023 **Fields Research Fellowship**: *Grothendieck Descent Theory, Fibrations, and Factorizations*. Principal investigator: Fernando Lucatelli Nunes; participant: Walter Tholen.
- 2022 **Oberwolfach Leibniz Fellowship**: *Grothendieck Descent Theory, Galois Theory, and Presentations of Categorical Structures*. Principal investigator: Fernando Lucatelli Nunes; participants included Maria Manuel Clementino, Rui Prezado, Lurdes Sousa, and Matthijs Vákár.

SELECTED PARTICIPATION IN RESEARCH PROJECTS AND GRANTS

- 2025–present **ERC Starting Grant FoRECAST**: *Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy*. PI: Matthijs Vákár. Member of the formal-methods team; contributing to categorical foundations and principled algorithms, and co-supervising doctoral research.
- 2026 **DFG Project HOMBRe**: *Higher-Order Monad-Based Programming and Reasoning*. PI: Sergey Goncharov. Research Fellow contributing to the project’s work on higher-order programming and semantics.
- 2021–2024 **NWO Veni Grant VI.Veni.202.124**: *Correct Expressive Differential Programming: Laying the Mathematical Foundations for Tomorrow’s Machine Learning*. PI: Matthijs Vákár. Led the work on iterative CHAD and contributed correctness results for automatic differentiation with recursive types.
- 2020–2022 **Marie Skłodowska-Curie Grant**: *Principled AD Program Transformations*. PI: Matthijs Vákár. Developed correct reverse-mode automatic-differentiation transformations for expressive total languages.
- 2011 **CNPq group project**: *Nonlinear Analysis and Applications*, University of Brasília. Led the undergraduate subproject on topological dynamics and applications to number theory and authored two monographs.

PHD SUPERVISION

CURRENT PHD STUDENTS

- 2025–present **Néot Choserot**, PhD in Mathematics. Project: *Aspects of Two-Dimensional Universal Algebra*.
- 2026–present **Diogo Simm**, Department of Information and Computing Sciences, Utrecht University. Project: programming-language semantics, probabilistic programming, algebraic effects, and automatic differentiation. Co-supervised with Dr. Matthijs Vákár; funded by ERC FoRECAST.

FORMER PHD STUDENT

2020–2024 **Rui Rodrigues de Abreu Fernandes Prezado**, PhD in Mathematics. Thesis: *Some Aspects of Descent Theory and Applications*. Co-supervised with Prof. Maria Manuel Clementino; funded by FCT. PhD awarded summa cum laude on 8 January 2024.

GRADUATE SUPERVISION

MASTER'S STUDENTS

2026–2028 **Kate Birch**, Master's Programme in Pure Mathematics, University of Cape Town. Research area: category theory. Co-supervised with Prof. Thomas Meyer.

2024–2025 **Diogo Simm**, Master's Programme in Pure Mathematics, IMPA, Rio de Janeiro. Research areas: programming-language semantics, automatic differentiation, and category theory. Co-supervised with Prof. Henrique Bursztyn and Dr. Matthijs Vákár; funded by CNPq. Degree awarded with distinction (Grade A).

SHORT-TERM MASTER'S PROJECT SUPERVISION

2025 **Scientific Methods in Computing Science (Programming Languages)**, Utrecht University. Supervised multiple small-group research projects on symbolic and automatic differentiation, program transformations, and categorical semantics. Representative projects included:

- *Calculating the Symbolic Derivative Using Automatic Differentiation*;
- *Lambda-Set Specialisation for Haskell*;
- *Dual-Numbers Automatic Differentiation with Inductive Types*.

UNDERGRADUATE SUPERVISION

THESIS AND CAPSTONE SUPERVISION

2023–2024 **BSc Computer Science Capstone Project**, Utrecht University. Supervised Céline Touber, Oscar Mensink, Kasper Peeters, Nigel Strachan, Joost Laan, Jippe Hoogeveen, Gust Verhees, Nasr Antar, and Artur Salek. The team designed and implemented a web-based optimisation system for identifying collaboration matches within organisations. Client: Dr. Ioanna Lykourantzou.

UNDERGRADUATE SOFTWARE PROJECTS

2021–2022 **Game Technology**, Utrecht University. Supervised nine undergraduate students in three teams designing and developing games in C#.

TEACHING, CURRICULUM DEVELOPMENT, AND PEDAGOGICAL LEADERSHIP

COURSE DESIGN AND CURRICULUM DEVELOPMENT

2026 **Object-Oriented Programming**, BSc Mathematics. Designed the syllabus, learning outcomes, assessment structure, programming-language orientation, and technical content, emphasising abstraction, modularity, software design, and the conceptual foundations of programming practice.

2026 Contributed to revision of the **Programming Methods I and II** sequence, aligning its objectives and methodology with the new Object-Oriented Programming course and strengthening the curricular foundation for programming, software development, and computational mathematics.

SELECTED TEACHING EXPERIENCE

SCHEDULED TEACHING, 2026–2027

2026–2027 Main lecturer: **Object-Oriented Programming**, BSc Mathematics and Computer Engineering.

2026–2027 Main lecturer: **Programming Methods I**, BSc Mathematics and Computer Engineering.

2026–2027	Main lecturer: Mathematical Analysis I , programmes in Mathematics, Computer Engineering, and Artificial Intelligence and Data Science.
2026–2027	Main lecturer: Introduction to Linear Algebra , programmes in Mathematics, Computer Engineering, and Artificial Intelligence and Data Science.
2026–2027	Main lecturer: Categories in Algebra and Topology , joint PhD programme in Mathematics. Topics include enriched categories, monads, and generalised multicategories.

COMPLETED AND CONTINUING TEACHING

2025–2026	Mathematical Analysis II , BSc Computer Engineering and Artificial Intelligence and Data Science; main lecturer.
2025–2026	Categories in Algebra and Topology , joint doctoral programme in Mathematics; main lecturer. Topics included adjunctions, Yoneda theory, monads, monoidal functors, and categorical logical relations.
2020–2025	Utrecht University , Department of Information and Computing Sciences; graduate teaching assistant. Courses and activities included Functional Programming, Logic for Computer Science, Graphics for Computer Science, BSc capstone supervision, and Game Technology project coaching. Recipient of the Utrecht University Distinguished Teaching Award (2023) .
2023–2024	Logic for Computer Science , Utrecht University, BSc Computer Science; main lecturer, co-taught with Dr. Wouter Swierstra. Topics included inference systems, natural deduction, semantics, and Hoare logic.
2019–2020	Categories in Algebra and Topology , joint doctoral programme in Mathematics; main lecturer.
2018–2019	Topology and Linear Analysis ; Invited Assistant Professor, co-taught with Prof. Maria Manuel Clementino.
2017–2020	Delfos Project ; advanced mathematics instruction for secondary-school students preparing for national and international competitions.
2008–2012	University of Brasília ; undergraduate teaching assistant in differential and integral calculus, real analysis, linear algebra, and Algebra I.
2006–2008	Curso Dínatos , Brasília; secondary mathematics teacher for students preparing for national university entrance examinations.
2005–2006	EPCAR, Brazilian Air Force ; student teaching assistant in mathematics.

INVITED AND PLENARY LECTURES

02/07/2026	CatAlg 2026 , UniMont, Edolo, Italy; invited lecture: <i>Beyond Mac Lane Strictification</i> (joint work with Paul B. Levy).
Online, 21/04/2026	Itaca Fest 2026 , invited lecture: <i>A Generalization of the Eilenberg–Moore Factorization</i> .
Messina, 09/2026	2nd Joint Meeting Brazil–Italy in Mathematics , University of Messina, Italy; forthcoming invited lecture on strictification.
Lorentz Center, Leiden, 2022	New Challenges in Programming Language Semantics , invited lecture: <i>A Categorical Approach to Semantic Logical Relations for Partial Features and Recursive Types</i> .
University of the Azores, 2018	International Conference on Category Theory 2018 , plenary lecture: <i>Aspects of Descent via Bilimits</i> .

SELECTED INVITED AND REFEREED TALKS

Leiden University, 2024	109th Peripatetic Seminar on Sheaves and Logic: <i>The Semantic Lax Descent Factorization of a Functor</i> .
Santiago de Compostela, 2024	International Conference on Category Theory 2024: <i>Doubly-Infinitary Distributive Categories</i> .
2024	SUMTOPO 2024, Topology and Logic in Computer Science: <i>Homomorphic Reverse Differentiation for Partial Features</i> .

2024	38th Summer Conference on Topology and its Applications: <i>Free Distributive and Extensive Categories</i> .
2023	XIV Portuguese Category Seminar: <i>Freely Generated Bicartesian Categories and Exponentials</i> .
UCLouvain, 2023	International Conference on Category Theory 2023: <i>Spanning the Tale of “Monades et descente”</i> .
2022	XIII Portuguese Category Seminar: <i>CHAD: Elementary Categorical Aspects of Automatic Differentiation</i> .
University of Primorska, 2021	8th European Congress of Mathematics: <i>Eilenberg–Moore, Kleisli and Descent Factorizations</i> .
2019	XII Portuguese Category Seminar: <i>Lax Comma 2-Categories</i> .
University of Edinburgh, 2019	International Conference on Category Theory 2019: <i>Descent and Monadicity</i> .
Stellenbosch University, 2019	Categorical Structures in Algebra, Topology and Logic: <i>Eilenberg–Moore, Kleisli and Descent Factorization</i> .
Masaryk University, Brno, 2018	103rd Peripatetic Seminar on Sheaves and Logic: <i>Biadjoint Triangles and Applications</i> .
Santiago de Compostela, 2018	102nd Peripatetic Seminar on Sheaves and Logic: <i>Non-Canonical Isomorphisms</i> .
UCLouvain, 2017	5th Workshop on Categorical Methods in Non-Abelian Algebra: <i>Biadjoint Triangles</i> .
2017	XI Portuguese Category Seminar: <i>Freely Generated n-Categories and Coinserter</i> .
2016	4th Workshop on Categorical Methods in Non-Abelian Algebra: <i>Commutativity of Bilimits and Descent</i> .
University of Aveiro, 2015	International Conference on Category Theory 2015: <i>Kan Extensions and Descent Theory</i> .

INVITED RESEARCH SEMINARS

University of Birmingham, 2026	Theory of Computation Seminar: <i>Eilenberg–Moore, Kleisli, and Descent Factorizations</i> .
IMPA, 2025	Algebra Seminar: <i>Biadjoint Triangles and Coherence</i> .
University of Cape Town, 2025	Category Theory Seminar: <i>Bifibrations, Spans, and the Bénabou–Roubaud Theorem</i> .
Stellenbosch University, 2025	Category Theory Seminar: <i>Two-Dimensional Monad Theory, Pseudodistributive Laws, and Doubly-Infinity Distributive and Extensive Categories as Pseudoalgebras</i> .
University of Cape Town, 2025	Category Theory Seminar: <i>Two-Dimensional Monad Theory and Free Extensive Completions</i> .
2024	Algebra, Logic and Topology Seminar: <i>Unraveling the Iterative CHAD</i> .
2022	Algebra, Logic and Topology Seminar: <i>Descent for Split Fibrations</i> .
Utrecht University, 2021	Software Technology Group Seminar: <i>Automatic Differentiation of Expressive Total Functional Languages</i> .
2019	Category Theory Seminar: <i>On Monadicity and Descent</i> .
2019	Algebra, Logic and Topology Seminar: <i>Functorial Semantics and Descent II</i> .
2018	Algebra, Logic and Topology Seminar: <i>Functorial Semantics and Descent</i> .
2018	Algebra, Logic and Topology Seminar: <i>Biadjoint Triangles and Applications</i> .
2018	Algebra, Logic and Topology Seminar: <i>Towards Semi-Left-Exactness and Janelidze–Galois Theory in the Lax-Idempotent Context</i> .
2016	Algebra, Logic and Topology Seminar: <i>The Deficiency of Categories</i> .

2015	Algebra, Logic and Topology Seminar: <i>Biadjoint Triangles, Descent and Coherence</i> .
University of Porto, 2016	Research Seminar Programme: <i>Kan Construction of Adjunctions</i> .
2016	Research Seminar Programme: <i>Homotopy Excision</i> .
2013	Joint PhD Programme Seminar: <i>Every Prespectrum Represents a Homology Theory</i> .
University of Brasília, 2010	Mathematics Seminar: <i>Topological Dynamics and Applications in Number Theory</i> (in Portuguese).
University of Brasília, 2010	Mathematics Seminar: <i>A Dynamical Proof of Van der Waerden's Theorem</i> (in Portuguese).

SELECTED WORKSHOP TALK

LAFI 2024, London	<i>Homomorphic Reverse Differentiation of Iteration</i> , joint work with Gordon Plotkin and Matthijs Vákár; Tenth Workshop on Languages for Inference at POPL.
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EVALUATION COMMITTEES

RECRUITMENT

2025–2026	Member and adviser, hiring committee for three PhD positions funded by the ERC Starting Grant FoRECAST (<i>Formalised Reasoning about Expectations: Composable, Automated, Speedy, Trustworthy</i>).
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DOCTORAL EXAMINATION COMMITTEES

2024	PhD defence committee, joint doctoral programme in Mathematics. Candidate: Rui Rodrigues de Abreu Fernandes Prezado; thesis: <i>Some Aspects of Descent Theory and Applications</i> ; role: supervisor and examiner.
2023	PhD defence committee, joint doctoral programme in Mathematics. Candidate: Carlos Miguel Alves Fitas; thesis: <i>On Lax Idempotent Monads in Topology</i> ; role: main examiner and opponent.

DOCTORAL PROJECT EVALUATION

Member of doctoral project evaluation committees in a joint doctoral programme in Mathematics, including evaluation of research proposals in category theory, topology, lax-idempotent monads, and Grothendieck descent theory.

OTHER ACADEMIC EVALUATION

Evaluation of BSc theses in Artificial Intelligence, Utrecht University, The Netherlands.

ACADEMIC SERVICE AND GOVERNANCE

GOVERNANCE

2023–2025	Member, Informatics Research Advisory Committee (IRAC), Department of Information and Computing Sciences, Utrecht University.
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ORGANISATION OF SCIENTIFIC EVENTS AND SEMINARS

2026–present	Co-organiser, Mathematics Colloquium.
2026–present	Organiser, Algebra, Logic and Topology Seminar.
2025–present	Organiser, Online Algebra, Logic and Topology Seminar.
2019–2023	Organiser, Algebra, Logic and Topology Seminar.
2022	Organising Committee Member, XIII Portuguese Category Seminar.
2019	Organising Committee Member, XII Portuguese Category Seminar.
2018	Organiser, Workshop on Algebra, Logic and Topology.
2018	Organiser, <i>Logic for Children</i> workshop, UNILOG 2018, Vichy, France.

OUTREACH AND PUBLIC ENGAGEMENT

2026–present	CLASS – Category, Logic, Algebra and Semantics Seminar. Founder and organiser of an international seminar for graduate students and researchers across pure mathematics and computer science, connecting participants in Portugal, South Africa, the United Kingdom, the Netherlands, China, Brazil, and Germany.
2017–2020	Delfos Project. Contributor to a mathematics-education programme for secondary-school students preparing for national and international competitions; Portuguese representative to the International Mathematical Olympiad in 2020.
2008–2012	Scientific Initiation (CNPq), Brazil. Undergraduate member of the research group <i>Teoria de Lie e Dinâmica</i> ; contributor to the outreach blog https://liedinamica.wordpress.com/ ; author of educational monographs in Portuguese on Lie groups, topology for dynamical systems, and applications of topological dynamics to number theory.
2011–2012	Brazilian Society for Scientific Progress (SBPC). Presented popular-science posters at national conferences on game theory, homotopy theory, category theory, topology, and dynamical systems; contributions published in the conference annals (in Portuguese).

EDITORIAL AND REFEREEING ACTIVITY

Referee for international journals and conferences in mathematics and computer science, including the ACM SIGPLAN Symposium on Principles of Programming Languages (POPL), *Advances in Mathematics*, *Applied Categorical Structures*, *Homology*, *Homotopy and Applications*, *Journal of Pure and Applied Algebra*, *Mathematical Structures in Computer Science*, *Order*, *Quaestiones Mathematicae*, *Theory and Applications of Categories*, and *Topology and its Applications*.